

INSTRUCTIONS

1. Mobile phones / any electronic gadgets (calculators etc) are strictly **PROHIBITED**.
2. Answers given without any justification will be awarded **zero** marks.

1. The optimal simplex tableau of an LPP **(P)**(with added variables s_1, s_2) is given below.

	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\tilde{\mathbf{a}}_3$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{b}$
$\mathbf{c}_j - \mathbf{z}_j$	0	-11	-1	0	-2	
x_1	1	2	-1	0	1	2
s_1	0	1	2	1	-1	0

- [3^{pnts.}] (a) Give an optimal BFS of the Dual of **(P)**. Does the Dual have a unique optimal solution? If yes then justify. If no then is the optimal solution set of the Dual unbounded? Justify.

Soln: An optimal BFS is $[0, 2, 0, 11, 1]^T$. marks[1]

Since the Primal has a degenerate optimal BFS and the corresponding optimal BFS of the dual is nondegenerate (since the $c_j - z_j$ values corresponding to the nonbasic variables are all nonzero in the primal's table which gives the values of the basic variables in the corresponding BFS of the Dual) and the $c_j - z_j$ value corresponding to the nonbasic variable y_1 of the Dual is zero, we can get multiple optimal solutions of the Dual by entering that variable at a positive value. So the dual has infinitely many optimal solutions but the optimal solution set of the Dual is bounded since there is no non negative row in the main simplex table(which gives a direction of the Dual) in any of the tables (corresponding to different Basis) of the optimal BFS given above. marks[2]

- [4^{pnts.}] (b) If the constraint $2x_1 - x_2 + x_3 \leq 3$ is added to **(P)** then find the optimal solution of the **new (P)** or conclude that the new **(P)** does not have a feasible solution, by using the Dual Simplex algorithm.

Soln: The first table after adding the constraint is

	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\tilde{\mathbf{a}}_3$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\mathbf{c}_j - \mathbf{z}_j$	0	-11	-1	0	-2	0	
x_1	1	2	-1	0	1	0	2
s_1	0	1	2	1	-1	0	0
s_3	0	-5	3	0	-2	1	-1

s_3 is the leaving variable and s_2 is the entering variable.

The next table is

	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\tilde{\mathbf{a}}_3$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\mathbf{c}_j - \mathbf{z}_j$	0	-6	-4	0	0	-1	
x_1	1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	$\frac{1}{2}$	$\frac{3}{2}$
s_1	0	$\frac{7}{2}$	$\frac{1}{2}$	1	0	$-\frac{1}{2}$	$\frac{1}{2}$
s_2	0	$\frac{5}{2}$	$-\frac{3}{2}$	0	1	$-\frac{1}{2}$	$\frac{1}{2}$

The above table gives the optimal BFS of **(P)**. marks[4]

[2pnts.]

- (c) In general (for any problem) if a variable enters a basis in one iteration of the Dual Simplex algorithm then can it leave the basis in the next iteration? Justify.

Soln: If the s th variable and the r th variable is the entering variable and the leaving variable respectively, in the current iteration then according to the rule for the leaving variable in Dual simplex $(B^{-1}\mathbf{b})_r < 0$. Since the rule of the entering variable says that $\frac{c_s - z_s}{u_{rs}} = \min\{\frac{c_j - z_j}{u_{rj}} : u_{rj} < 0\}$, (for a maximization problem) so $u_{rs} < 0$.

So the pivot $u_{rs} < 0$. In the next table the r th row is divided by u_{rs} which makes the value of the entering variable as $\frac{(B^{-1}\mathbf{b})_r}{u_{rs}}$ which is > 0 , hence it cannot leave the basis in this new iteration.

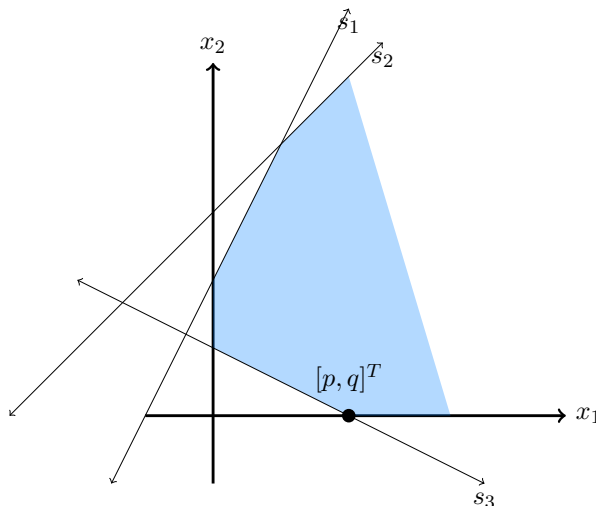
Aliter: This equivalent to a variable in the simplex table of the Dual leaving the basis and entering the basis in the next iteration which is not possible as discussed. You have to give all necessary justifications for this. marks[2]

2. Consider LPP (P) of the form,

$$\text{Minimize } \mathbf{c}^T \mathbf{x}$$

$$\text{subject to } A_{3 \times 2} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0},$$

with the shaded feasible region given below. Non negative variable s_i , $i = 1, 2, 3$ is added to the i th constraint of $A_{3 \times 2} \mathbf{x} \leq \mathbf{b}$ and the hyperplanes corresponding to each of the added variables s_i are marked accordingly in the picture given below.



[3pnts.]

- (a) Give the sign of all the entries in the **non basic** columns in the simplex table (not the $c_j - z_j$ values) of the **BFS** corresponding to $[p, q]^T$ with adequate justification.

Soln: The sign is given as follows:

	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{e}_3$	$B^{-1}\mathbf{b}$
$\mathbf{c}_j - \mathbf{z}_j$						
x_1		+			-	
s_1		+			-	
s_2		+			-	

You have to give correct justification as to why the above table is correct. If the argument is wrong/missing then even if the table is correct **no** marks are awarded for such answers.

[3pnts.]

- (b) If $[p, q]^T$ is optimal for (P) with basis matrix B and if b_3 (the third component of $\mathbf{b}$) is changed to b'_3 then will the new (P) always have an optimal solution? If yes then which are the possible optimal solutions of the new (P)? If no then justify.

Soln: The new problem will always have an optimal solution since however you change b_3 new (P) will always have a nonempty feasible region. Also since the feasible region of the Dual of the new (P) is same as the feasible region of the Dual of the old (P) which is non empty since (P) has optimal solution, so by duality theory new (P) has

an optimal solution. [1]

Consider the basis matrix B corresponding to $[p, q]^T$ such that in the corresponding simplex table the $c_j - z_j$ values are all nonnegative. Since $B^{-1}\mathbf{b}' \geq \mathbf{0}$ if the line is shifted to the right (or away from the origin) of the current position (where $B^{-1}\mathbf{b}'$ is $B^{-1}\mathbf{b}$ with b_3 replaced with b'_3) $[(B^{-1}\mathbf{b}')^T, 0]^T$ will be the new optimal solution (since the $c_j - z_j$ values are all nonnegative) which will be the extreme point given by the intersection of the lines corresponding to $s_3 = 0$ and $x_2 = 0$.

When the hyperplane is moving towards the origin or to the left of the current position then also $[(B^{-1}\mathbf{b}')^T, 0]^T$ will be the optimal solution (by the same argument) upto $b'_3 = 0$ when $\mathbf{0}$ will be the optimal solution, which is the extreme point corresponding to the lines $s_3 = 0$ and $x_2 = 0$. When it moves further to the left then since the feasible region remains the same as when $b'_3 = 0$ and the objective function also remains unchanged $\mathbf{0}$ is the optimal solution. [2] marks